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APUL

NEET FLIX

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BIOLOGY

1) Find out **incorrect** statement.

(1) The vegetative cell of pollen grain has large irregular shaped nucleus.
(2) Intine of pollen grain is made up of pectocellulose.
(3) Generative cell of pollen grain divides meiotically to give rise to the two male gametes.
(4) Each cell of the sporogenous tissue is capable of giving rise to a microspore tetrad.

2) Which one of the following statement is not correct?

(1) Pollen grains are released from anthers at 2-celled stage
(2) Primary sporogenous cell directly behaves as the megaspore mother cell
(3) Megaspore divides twice to form eight nucleated embryosac
(4) Egg and synergids always lie near the micropylar end of ovule

3) Read carefully the given statement-
Statement-I : In some species of flowering plants, the rewards in return to pollination are in providing safe places for pollinators to lay eggs.
Statement-II : Generally single ovule is present in each ovary in wind pollinated flowers.
Choose the correct answer.

(1) Both Statement-I and Statement-II are correct
(2) Both Statement-I and Statement-II are incorrect
(3) Statement-I is correct but Statement-II is incorrect
(4) Statement-I is incorrect but Statement-II is correct

4) **Statement - I :** Pollen grain are generally spherical measuring about 25 – 50 micrometers.
Statement - II : Pollen grain are well preserved because of presence of sporopollinin.

(1) Both statement-I and statement-II are incorrect.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect but statement-II is correct.
(4) Both statement-I and statement-II are correct.

5) **Statement-I :** Endosperm development precedes embryo development.
Statement-II : Embryo develops at the micropylar end of the embryo sac where the zygote is situated.

(1) Statement-I & Statement-II both are correct.
(2) Statement-I is correct & Statement-II is incorrect.
(3) Statement-I is incorrect & Statement-II is correct.

(4) Both Statement-I & Statement-II are incorrect.

6) **Statement-I :** All flowering plants show sexual reproduction.
Statement-II : Flowers are objects of asthetics, ornamental, social and religious value.

(1) Statement-I & Statement-II both are correct.
(2) Statement-I is correct & Statement-II is incorrect.
(3) Statement-I is incorrect & Statement-II is correct.
(4) Both Statement-I & Statement-II are incorrect.

7) Match a, b, c and d with i, ii, iii and iv:

(a)	Pollen tablets	(i)	Pollen viability for months
(b)	Rosaceae	(ii)	Pollen Allergy
(c)	Pollen bank	(iii)	Food supplement
(d)	*Parthenium*	(iv)	–196°C

Choose the correct match :
(1) a-i, b-iii, c-ii, d-iv
(2) a-iii, b-iv, c-i, d-ii
(3) a-iii, b-i, c-iv, d-ii
(4) a-ii, b-i, c-iv, d-iii

8)

(a) *Michelia*	(i) One ovule in each ovary
(b) *Papaver*	(ii) Many ovule in each ovary
(c) Papaya	(iii) Mono carpellary
(d) Paddy	(iv) Multicarpellary syncarpous
	(v) Multicarpellary, Apocarpous

Choose the correct match :
(1) a-i, b-ii, c-iii, d-iv
(2) a-v, b-iii, c-ii, d-i
(3) a-v, b-iv, c-ii, d-i
(4) a-iv, b-v, c-i, d-ii

9)

How many plants in the list given below are monocarpic :-

Wheat, Rice, Agave, Mustard, Carrot, Pea, Bamboo, Capsella.
(1) 7
(2) 8

(3) 4

(4) 3

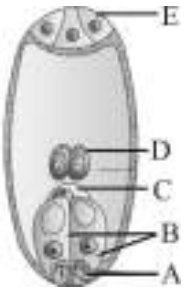
10) Maize, Date palm, Grass, Coconut, Cotton, *Eucalyptus*, Apple, Lemon, Papaya, Bamboo.
How many of the above plants have anemophily ?

(1) Four

(2) Five

(3) Six

(4) Seven

11) 

In above diagram which of the following is correct function of the corresponding labelling :-

(1) C - Formation of primary endosperm.

(2) E - Entry of pollen tube in embryosac.

(3) A - Guide the entry of pollen tube.

(4) D - Formation of zygote after fusing with one of the male gamete.

12)

Consider the following four statements (a-d) and select the option which includes all the correct ones :
(a) A multicarpellary apocarpous gynoecium is found in michelia
(b) Pollen robbers consume pollen or nector without bringing pollination.
(c) The transformation of ovules into seed and ovary into fruit proceed simultaneously

(d) The pollen grain represent the male gametophyte.

(1) a, b and c

(2) a, b, c and d

(3) b, c and d

(4) a, b and d

13)

Read the following four statements (A-D) and answer as asked next to them :-
(A) A typical anther is a unilobed structure consisting of two pollen sacs.
(B) Each cell of the sporogenous tissue is capable of giving rise to a microspore tetrad.
(C) Sporopollenin is one of the most resistant inorganic material known.
(D) The inner most layer of anther wall is tapetum, it nourishes the developing pollen grains.

How many of the above statements are correct?

(1) Four

(2) One

(3) Two

(4) Three

14) Read the statements carefully :-
1. Flowers are morphological and embryological marvel and the site of sexual reproduction.
2. The proximal end of the filament is attached to the thalamus or the petal of the flower.
3. The number and length of stamens are variable in flowers of different species.
4. A typical anther is bilobed and tetrasporangiate.
How many above statements are correct :-

(1) One

(2) Two

(3) Three

(4) Four

15) A flower has five stamens. Each stamen has tetrasporangiate anther. If 30 pollen mother cells are present in each microsporangium then how many pollen grains will be produced by this flower?

(1) 140

(2) 2400

(3) 600

(4) 4800

16) Consider the following four statements (A-D) and select the option which includes all the correct ones only :-
A. The portion of the embryonal axis above the level of cotyledons is the epicotyl which terminates with the plumule
B. Seeds of beet possesses remnant of nucellus called perisperm.
C. Parthenocarpy can be induced through the application of growth hormones.
D. Seeds have better adaptive strategies for dispersal to new habitats and help the species to colonise in other areas.

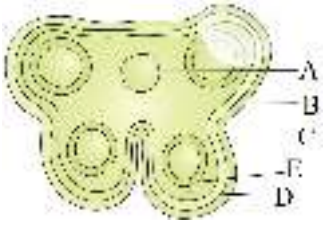
(1) Only statements A, B, C

(2) Only statements B, C, D

(3) Only statements A, B, D

(4) Statements A, B, C, D

17) Choose the correct function of the given structures :



(1) A - connect anther and filament ;B - Help in dehiscence of anther

(2) B - Protection ; E - Nutritive layer

(3) C - Dehiscence of anther ; D - Nutritive layer

(4) A - Protection ; C - Nutritive layer

18) How many times meiosis and mitosis required for production of embryo sac respectively?

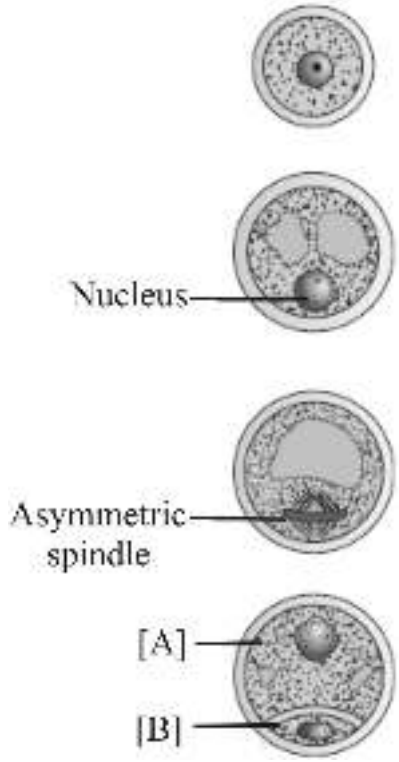
(1) 1, 3

(2) 1,2

(3) 1, 1

(4) 1, 4

19) Stages of a microspore maturing into a pollen grain are given in the diagrams. Identify labelled



cells [A] and [B] of given diagram :-

(1) [A] Generative cell, [B] Vegetative cell

(2) [A] Generative cell, [B] Tube cell

(3) [A] Vegetative cell, [B] Tube cell

(4) [A] Vegetative cell, [B] Generative cell

20) In some plants, pollen grains are shed at two celled condition. In such plants, the generative cell divides and forms.....during the growth of pollen tube on the stigma.

(1) Vegetative cell & male gamete

(2) Two female gametes

(3) One male gamete

(4) Two male gametes

21) In angiosperms following events occur during microgametogenesis :-
(a) Mitotic division in microspore.
(b) Meiotic division in miorospore mother cell.
(c) Formation of pollen grain.

(d) Formation of microspore.
(e) Formation of mature male gametophyte.
The correct sequence of event is :-

(1) b, a, d, e, c

(2) c, b, a, e, d

(3) b, d, a, c, e

(4) d, e, b, a, c

22)

Which of the following is correct?
(i) Male gametes in angiosperms are formed by the division of vegetative cell.
(ii) Hilum represents the junction between body of ovule and funicle.
(iii) Chasmogamous flowers produce assured seed set even in the absence of pollinators.

(iv) Residual, persistant nucellus is the perisperm.

(1) i, ii, iv

(2) i, ii, iii

(3) i, iii, iv

(4) ii, iv

23) Which of the following is wrongly matched?

(1) Pollen sacs – Microsporangia

(2) Stamen – Microsporophyll

(3) Carpel – Megasporophyll

(4) Pollen grain – Male gamete

24)

Read the following four statements (A-D) :-
(A) Functional megaspore develops into the male gametophyte.
(B) Polar nuclei are situated below the egg apparatus in the central cell.
(C) Male and female gametes of flowering plants are non-motile.
(D) The most reduced male gametophyte is found in angiosperms
How many of the above statements are right?

(1) Four

(2) One

(3) Two

(4) Three

25) If leaftip cells possess 28 chromosomes, the number of chromosomes in middle layer, generative cell and male gametes will be:

(1) 14, 28, 14

(2) 28, 14, 14

(3) 14, 14, 28

(4) 28, 28, 14

26) The alerone layer in maize grain is specially rich in & its ploidy is respectivelty :-

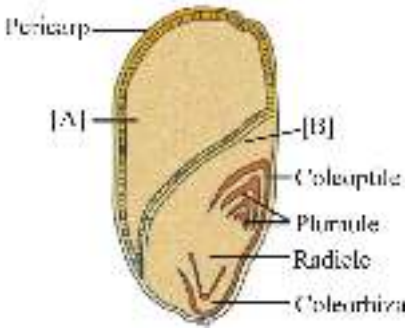
(1) Proteins, 3n

(2) Proteins, 2n

(3) Starch , 3n

(4) Lipids, 3n

27) L.S. of a maize grain is given below. Identify labelled structures [A] & [B] :-



(1) [A] Embryo , [B] Scutellum

(2) [A] Scutellum , [B] Embryo

(3) [A] Endosperm, [B] Scutellum

(4) [A] Scutellum , [B] Endosperm

28) The ploidy levels of the cells of the nucellus, MMC, the functional megaspore and female gametopyte are :-

(1) 3N, N, 2N, N

(2) N, N, 2N, N

(3) 2N, 2N, N, N

(4) N, 2N, 2N, N

29) Which type of flowers can produce seeds even on the absence of pollinators ?

(1) Chasmogamous flower

(2) Unisexual flower

(3) Cleistogamous flowers

(4) Bisexual flower

30) Arrange the following wall layer of anther from innerside to outside :-
(a) Tapetum (b) Endothecium
(c) Epidermis (d) Middle layer

(1) (a), (d), (b), (c)

(2) (c), (b), (d), (a)

(3) (b), (d), (b), (c)

(4) (b), (d), (a), (c)

31) Pollen grain represents

(1) Female gametophyte
(2) Male gametophyte
(3) Sporophyte
(4) Anther

32) The arrangement of the nuclei in a normal embryosac in the dicot plants is

(1) 2 + 4 + 2
(2) 3 + 2 + 3
(3) 2 + 3 + 3
(4) 3 + 3 + 2

33) Which one is female gametophyte

(1) Embryo
(2) Embryosac
(3) Endosperm
(4) Pistil

34) Which one of the following pairs of plant structures has haploid number of chromosomes ?

(1) Nucellus and antipodal cells
(2) Egg nucleus and secondary nucleus
(3) Megaspore mother cell and antipodal cells
(4) Egg cell and antipodal cells

35) An example of a seed with endosperm, perisperm, and caruncle is :-

(1) Castor
(2) Cotton
(3) Coffee
(4) Lily

36) The scutellum observed in a grain of wheat or maize is comparable to which part of the seed in other monocotyledons ?

(1) Plumule
(2) Cotyledon
(3) Endosperm
(4) Aleurone layer

37) In a pollen grain the small cell is spindle shaped, with dense cytoplasm is :-

(1) Vegetative cell
(2) Generative cell
(3) Tube cell
(4) All

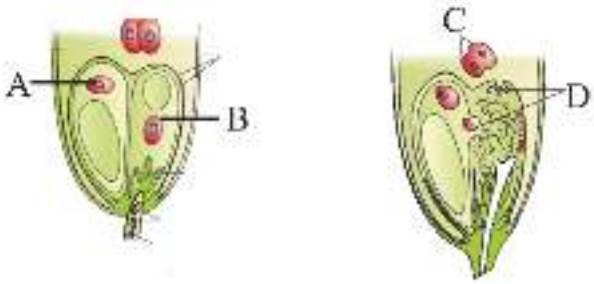
38) Apomictic embryos in citrus arise from :

(1) Diploid egg,
(2) Synergids
(3) Maternal sporophytic tissue in ovule
(4) Antipodal cells

39) How many pollen sacs are present in a mature anther

(1) 4
(2) 1
(3) 3
(4) 2

40) Choose the **correct** match with respect to given diagrams showing events of sexual reproduction



:-

(1) A - Synergid, B - Degenerating synergid
(2) C - Polar nuclei, D - Male gametes
(3) A - Synergid, D - Male gametes
(4) C - Polar nuclei, D - Synergid nucleus

41) Following diagram represents fertilized embryo sac. Identify labelled structure [A] of this

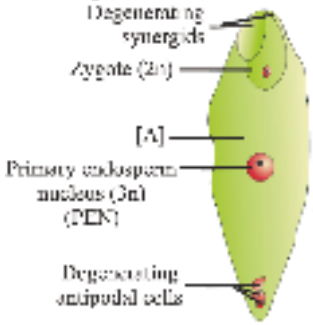
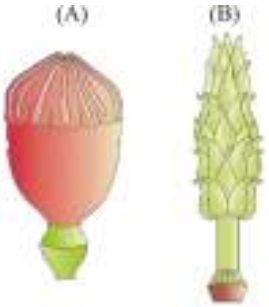


diagram:-

(1) Embryo
(2) Perisperm
(3) Primary endosperm cell

(4) Central cell



42) Identify the type of ovary in the following diagram (A) and (B) :-

(1) (A) Monocarpellary, syncarpous (B) Multicarpellary, apocarpous
(2) (A) Multicarpellary, apocarpous (B) Multicarpellary, syncarpous
(3) (A) Multicarpellary, syncarpous (B) Multicarpellary, apocarpous
(4) (A) Monocarpellary, apocarpous (B) Multicarpellary, apocarpous

43) Which of the following structure is shown in the given diagram :-



(1) Pollinium
(2) Tapetal cells
(3) Embryo-sac
(4) Pollen grain tetrad

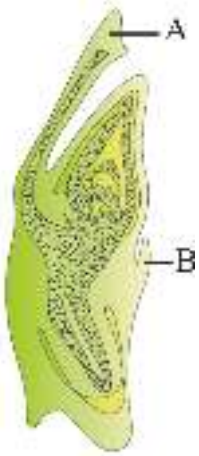
44)

The given plant is pollinated by :-



(1) Wind
(2) Water
(3) Insect
(4) Bird

45) Given below is digramatic figure of embryo of grass. Identify the labelled parts & select right



option about them.

	Column-I	Column-II
(1)	Coleoptile	Epiblast
(2)	Scutellum	Coleoptile
(3)	Epiblast	Coleoptile
(4)	Scutellum	Epiblast

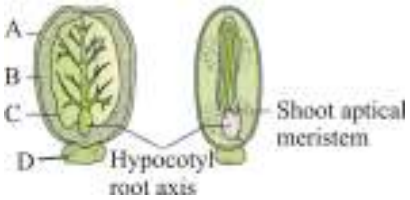
(1) 1

(2) 2

(3) 3

(4) 4

46) The diagrams are related to castor seeds. Identify A, B, C and D, respectively.



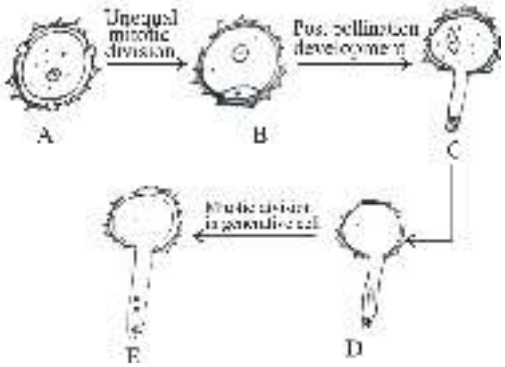
(1) Endosperm, seed coat, cotyledon and caruncle

(2) Seed coat, endosperm, caruncle and cotyledon

(3) Seed coat, cotyledon, endosperm and caruncle

(4) Seed coat, endosperm, cotyledon and caruncle

47) The figure given below represents the various steps of microgametogenesis. Select the option giving correct identification together with what it represents?



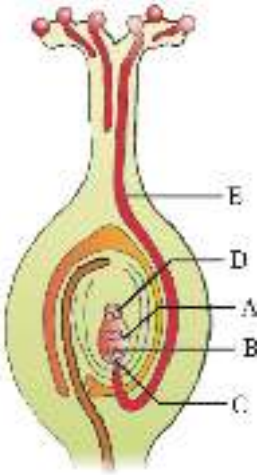
(1) (C) represents mature male gametophyte

(2) (B) Represents mature pollen grain

(3) (A) Represents mature male gametophyte

(4) (B) Represents mature male gametophyte

48) Given figure is longitudinal section of a flower showing growth of pollen tube. In which labelled parts A, B and C are respectively:



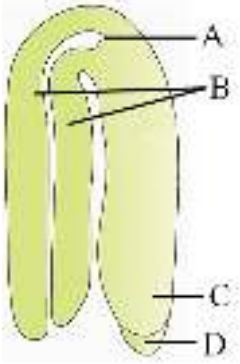
(1) Egg cell, Polar nuclei, Antipodal cell

(2) Polar nuclei, Egg cell, synergid

(3) Egg cell, Antipodal cell, synergid

(4) Pollen tube, synergid, Egg cell

49) Given below is the diagrammatic new of a typical dicot embryo. Identify the parts labelled A, B, C and D and select the right option about them.



Option

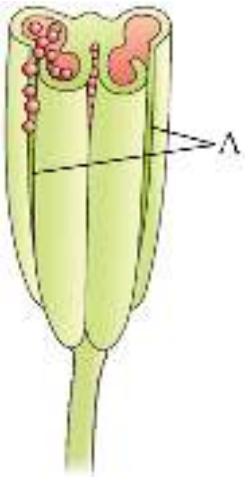
	A	B	C	D
1.	Radicle	Plumule	Cotyledons	Root cap
2.	Plumule	Cotyledons	Root cap	Radicle
3.	Plumule	Cotyledons	Radicle	Root cap
4.	Radicle	Cotyledons	Plumule	Root cap

(1) 1

(2) 2

(3) 3

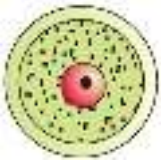
(4) 4



50) Identify A in the given diagram-

(1) Filament
(2) Anther
(3) Line of Dehiscence
(4) Pollen grains

51) Few diagram regarding the development of pollen grain are given below, arrange these diagrams in correct order and find out the correct option from options given below :-

(A) 
(B) 
(C) 
(D) 
(E) 

(1) A, B, C, D, E
(2) A, B, D, C, E
(3) A, C, D, B, E
(4) A, C, B, D, E

52) *Zostera* is an angiospermic plant and their pollen grains transfer with the help of :-

(1) Wind
(2) Water
(3) Birds
(4) Insects

53) Cleistogamous flowers are present in:

(1) *Commelina*
(2) Mustard
(3) *Mirabilis*
(4) *Catharanthus*

54) Plants with ovaries having only one or a few ovules are generally pollinated by :

(1) Birds
(2) Wind
(3) Bees
(4) Butterflies

55) Functionally cross pollination but genetically autogamy (self pollination) is ?

(1) Xenogamy
(2) Geitonogamy
(3) Autogamy
(4) Both (1) and (2)

56) How many plants in the list given below show hydrophily ?
Water hyacinth, water lily, *Vallisneria, Hydrilla, Zostera,* maize, *Yucca, Amorphophallus*

(1) Six
(2) Three
(3) Four
(4) Five

57) **Assertion :** Pollination by water is quite rare in flowering plants.
Reason : Pollination by wind is more common amongst biotic pollinators.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

58) **Assertion :-** When embryo sac develops from Megaspore mother cell without meiosis it is called Diplospory.
Reason :- Fruits are formed without seeds in parthenocarpy.

(1) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
(2) Assertion is correct but Reason is not correct.
(3) Assertion is not correct but Reason is correct.
(4) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.

59) **Assertion :-** Pollen grain can withstand high temperatures and strong acids and alkali.
Reason :- Exine of pollen grain is made of sporopollenin, which is most resistant organic material.

(1) Both Assertion & Reason are True and the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

60) **Assertion :** Filliform apparatus guides the pollen tube into the synergids.
Reason : It is special cellular thickening at micropylar tip that secrete chemotropic stimulus.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

ANSWER KEYS

BIOLOGY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	**3**	**3**	**1**	**4**	**1**	**1**	**3**	**3**	**2**	**3**	**3**	**2**	**3**	**4**	**2**	**4**	**2**	**1**	**4**	**4**
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	**3**	**4**	**4**	**4**	**2**	**1**	**3**	**3**	**3**	**1**	**2**	**2**	**2**	**4**	**1**	**2**	**2**	**3**	**4**	**2**
Q.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	**3**	**3**	**4**	**1**	**4**	**4**	**2**	**2**	**3**	**3**	**2**	**2**	**1**	**2**	**2**	**2**	**3**	**1**	**1**	**1**

SOLUTIONS

BIOLOGY

1) **Question Explanation :**
The question asks to identify the incorrect statement among the four options provided regarding the structure and development of pollen grains in flowering plants.
Concept:
This question is based on Structure Of Pollen Grain.
Explanation: Generative cell of pollen grain divides meiotically to give rise to the two male gametes. **- Incorrect**
The generative cell divides mitotically, not meiotically, to form two male gametes.
Final answer: Generative cell of pollen grain divides meiotically to give rise to the two male gametes.

2) NCERT (XII) Pg. # 22,26

3) NCERT, Pg. # 13, 14

4) NCERT-XI Pg. # 23

5) NCERT XII Pg # 34, 35

6) NCERT XII Pg # 19

7) **Question Explanation:** We have to match uses/examples of pollen with the correct statements.

Concept: Pollen – viability, storage, uses

Solution: Pollen tablets → used as food supplements (rich in nutrients, proteins).
Family Rosaceae → pollen grains viable for months.
Pollen banks → store pollen at –196°C in liquid nitrogen.
Parthenium (carrot grass) → causes pollen allergy.

Final answer: Option 3: a–iii, b–i, c–iv, d–ii

8)

Explain Question : The question is about matching plants to their ovary or carpel structure/ovule number.

Concept : Gynoecium and Ovule Number.

Solution:

A. (a) Michelia → Multicarpellary, Apocarpous (v).

B. (b) Papaver → Multicarpellary, Syncarpous (iv).

C. (c) Papaya → Has many ovules in each ovary (ii).

D. (d) Paddy → Has One ovule in each ovary (i).

The correct match is a-v, b-iv, c-ii, d-i.

Final Answer : Option (3)

9)

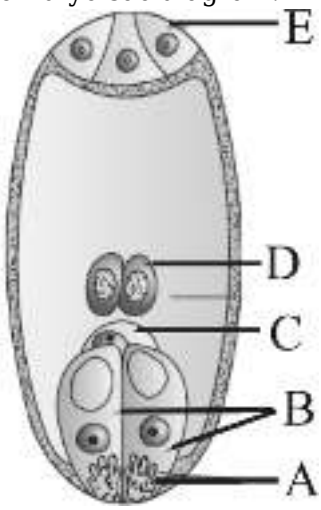
NCERT XII, Pg. # 9

10)

NCERT XII, Pg. # 14

11)

Question Explanation: Identifying the correct function for a labeled cell/structure in the embryo sac diagram.



Concept : Embryo Sac Structure and Function.

Solution :
The correct answer is option 3: A – Guide the entry of pollen tube.
Explanation: The diagram shows the 7-celled, 8-nucleate embryo sac.

– A is the Filiform Apparatus, located on the synergid cells. Its primary function is to secrete chemical signals that guide the pollen tube to the egg apparatus.
– C is the Egg Cell (the female gamete) which fuses with a male gamete to form the zygote.
– D represents the Polar Nuclei, which fuse with the other male gamete to form the Primary

Endosperm Nucleus.
– B is a synergid cell, which assists in fertilization and guides the pollen tube.
– E is three Antipodal cells.

Final Answer : Option (3) A - Guide the entry of pollen tube.

12) NCERT (XII) Pg. # 31

13) NCERT XII, Page # 21, 22, 23
B & E statements are correct

14)

NCERT XII, Pg. # 3

15)

NCERT XII, Pg. # 6

16)

NCERT XII, Pg. # 18

17)

NCERT XIIth Pg.#22 Figure- 2.3(a)

18)

NCERT XII, Pg # 10, 11

19) NCERT-XII Pg. # 7(E), 24(H) figure 1.5(A)

20) NCERT XII Pg # 17

21)

Explain Question : The question asks for the correct sequence of events during microgametogenesis in angiosperms.
Concept : microsporogenesis in angiousperms.
Solution :
b – MMC undergoes meiosis.
d – Microspores are formed.
a – Each microspore undergoes mitosis.
c – Pollen grain formation.
e – Mature male gametophyte formation.

Final Answer : option (3)

22)

Question Explanation: The question asks to identify the correct statements related to reproduction in flowering plants
Concept: This question is based on sexual reproduction in flowering plants (angiosperms)
Solution:
(ii) Hulum represents the junction between body of ovule and funicle. This statement is correct. The hilum is the point of attachment where the funicle stalk connects the ovule to the placenta, effectively marking the junction between the body of the ovule and the funicle.
(iv) Residual, persistant nucellus is the perisperm. This statement is correct. In some seeds, the nuclellus (nutritive tissue surrounding the embryo sac) persists as a thin, dry, and often papery layer called the perisperm, providing nourishment to the developing embryo.
Final Answer: (4)

23) **Question Explanation :** You are being asked to find the incorrect match between structures and their biological identity or function in the reproductive system of flowering plants.
Concept : Pre-Fertilization Events: Male Reproductive Whorl -Androecium To Development Of Male Gametophyte
Solution :
A pollen grain is not the male gamete, but a male gametophyte.
• It produce the male Male gamete after germination.
• So, equating pollen grain = male gamete is incorrect.
Final Answer : option (4)

24)

NCERT XII pg. 26-27

25) **Question Explanation:** The question is asking about determining the chromosome number at different stages of male gametophyte development starting from somatic cells.

Concept: Androecium

Solution:

Leaf tip cells are somatic cells → diploid (2n = 28) and Middle layer / Pollen mother cell (PMC) → arises from somatic tissue → diploid (2n = 28).

Generative cell → formed from haploid microspore after meiosis → haploid (n = 14).

Male gametes → formed by mitotic division of generative cell → remain haploid (n = 14).

Final answer: 2

26)

NCERT XII, Pg. # 21

27)

Question Explanation: Identifying the labeled parts A and B in the longitudinal section of a maize grain.

Concept: Monocot Seed Structure

Solution: The diagram shows the maize grain (caryopsis):
[A] points to the large food storage region, the Endosperm.
[B] points to the single, shield-shaped cotyledon, the Scutellum.

Final Answer: option (3)

28)

NCERT XII Pg 26

29) NCERT XII, Pg # 28, Fig. 2.9

30) NCERT XII, Pg # 22, Fig. 2.3 (b)

31)

NCERT XII, Pg. # 7

32)

NCERT XII, Pg. # 10

33)

NCERT XII, Pg. # 10

34)

NCERT XII, Pg. # 10

35)

NCERT XII, Pg. # 21

36)

NCERT XII, Pg. # 19 Fig. 1.14

37)

NCERT XII, Pg. # 7

38)

NCERT XII, Pg. # 22

39)

NCERT XII, Pg. # 6

40) **Question Explanation:** Identifying the labeled parts in diagrams illustrating pollen tube entry and post-fertilization changes.

Concept: Fertilization in Angiosperms

Solution:
In the left diagram → A is Egg nucleus, B is synergid.
In the right diagram → C is Polar nuclei, D is Male gametes entering the embryo sac.

Final Answer: Option 2. C - Polar nuclei, D - Male gametes

41)

NCERT-XII, Pg. # 18

42)

Question Explanation: Identifying the types of multicarpellary gynoecium shown in the diagrams of Papaver (A) and Michelia (B).

Concept: Gynoecium Types

Solution:

A. (A) Papaver has multiple carpels that are fused → Multicarpellary, syncarpous.

B. (B) Michelia has multiple carpels that are free → Multicarpellary, apocarpous.

Final Answer: (A) Multicarpellary, syncarpous (B) Multicarpellary, apocarpous

43) NCERT (XII) Pg # 23(E), 24(H)

44)

NCERT-XII Pg. # 13

45)

Question Explanation: Identifying the labeled parts A and B in the diagram of a grass embryo.

Concept : Monocot Embryo Structure.

Solution: The figure shows the longitudinal section of a grass (monocot) embryo:
A points to the protective sheath around the embryonic shoot, the Coleoptile.
B points to the small remnant flap opposite the scutellum, the Epiblast.

Final Answer : option (1)

46)

NCERT, Pg. # 21

47)

NCERT XII, Pg. # 7

48)

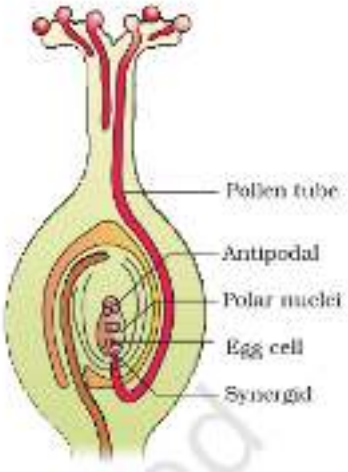
Question Explanation: Question is asking about, identifying the labeled parts A, B, and C in the given diagram showing pollen tube growth.

Concept: Fertilisation

Solution:

The given diagram showing L.S. of pistil showing path of pollen tube growth and labelled parts A, B and C are -

A-Polar nuclei, B-Egg cell,C- synergid



Final answer: 2

49)

Question Explanation: Identifying the labeled parts A, B, C, and D of the typical dicot embryo.

Concept: Dicot Embryo Structure

Solution: The diagram shows a dicotyledonous embryo (e.g., Pea or Bean):

A. A points to the embryonic shoot tip, the Plumule.

B. B points to the large food-storing leaves, the Cotyledons.

C. C points to the embryonic root tip, the Radicle.

D. D points to the protective tip covering the radicle, the Root cap.

Final Answer: 3

50) NCERT, Pg # 5

51)

Question Explanation: Arranging the stages of pollen development in the correct sequence.

Concept: Development of microspore and male gametophyte

Solution: The correct sequence of microspore development is -
(A) Enlarged view of a pollen grain tetrad.
(B) Microspore separates, nucleus is central.
(D) Stage of a microspore maturing into a pollen grain
(C) Asymmetrical spindle
(E) Mitotic division forms the Vegetative (large) and Generative (small) cell.

Final answer: Option 2

52)

NCERT XII Page # 13

53)

NCERT XII, Pg. # 12, Fig. 1.9

54) **Question Explanation :**
The question asks to identify the most likely pollinating agent for plants that have an ovary containing only a single or a few ovules.

Concept:
This question is based on pollination.

Explanation : Plants with ovaries containing only one or a few ovules are typically pollinated by wind. This is because wind pollination is less precise than animal pollination, and a large number of ovules would be wasted if not fertilized.
Final answer : Option 2 Wind

55)

NCERT XIIth Pg.#28(E), 30(H)

56) NCERT Page # 31

57) NCERT XII Pg # 29

58)

NCERT XII, Pg. # 21

59)

NCERT-XII-Page No. # 23

60) NCERT Pg#33